# Measuring Error
## Numerical Methods

David Mayerich

**Scalable Tissue Imaging and Modeling (STIM) Laboratory**

**Department of Electrical and Computer Engineering**

**Cullen College of Engineering**

**University of Houston**

UNIVERSITY of HOUSTON | ECE

# Error Behavior

- Define error behavior using Big-O notation and error terms:

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + O\left(x^4\right) \qquad \text{where} \qquad O\left(x^n\right) \leq C\left|x^n\right|$$

- Bound differences between errors as parameters change:

$$\begin{array}{llll} x_1 = 0.50 & n_1 = 3 & \epsilon_1 \leq C\left|0.5\right|^3 & \epsilon_1 \leq 0.125C \\ x_2 = 0.75 & n_2 = 7 & \epsilon_2 \leq C\left|0.75\right|^4 & \epsilon_2 \leq 0.316C \end{array} \qquad \frac{\epsilon_1}{\epsilon_2} \approx 0.396$$

- We can calculate how error bounds change without precisely knowing $\epsilon_1$ or $\epsilon_2$
- More precise values for $\epsilon$ require approximating $C$ in some way:
  - theoretically using features of the function
  - practically by comparing the result to some ground truth

# Approximating Error Analytically

- Consider the Lagrange remainder in a finite Maclaurin series expansion:

$$f(x) = \sum_{n=0}^{k} \frac{f^{(n)}(0)}{n!}(x)^n + \boxed{\frac{f^{(k+1)}(z)}{(k+1)!}x^{k+1}}$$

$$\frac{f^{(k+1)}(z)}{(k+1)!}x^{k+1} = O\left(x^{k+1}\right)$$

$$\frac{f^{(k+1)}(z)}{(k+1)!}x^{k+1} = C\left|x\right|^{k+1}$$

$$C = \frac{f^{(k+1)}(z)}{(k+1)!} \quad \text{where } z \in [0, x]$$

- What is $\epsilon$ for $e^x$ using a 5th-order series?

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \frac{x^5}{120} + \epsilon$$

1. If $\frac{d}{dx}e^x = e^x$ then $f^{(d+1)}(z) = e^z$
2. $e^x$ is monotonically increasing, so the largest value for $z \in [0, x]$ is $z = x$
3. The numerator is **at most** $e^x$
4. The error **in the worst case** is $\epsilon = \frac{e^x}{6!}x^6$

# Approximating Error Practically

- Absolute error is the magnitude of the difference between true and calculated values:

$$\epsilon_a = |\epsilon| = |t(x) - c(x)|$$

- How do we get the true value $t(x)$?
- Once again, consider $e^x$ as a 5th-order Maclaurin series:

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \frac{x^5}{120} + \epsilon$$

```
float exp5(x){

    float ex = 1;                        // initialize with e^0
    float xp = 1;                        // initialize with x^0
    float fac = 1;                       // initialize 0!
    for(int i = 1; i <= 5; i++){
        xp *= x;
        fac *= i;
        ex += xp / fac;
    }
    return ex;
}
```

```
exp5(1.0f) = 2.716667
```

NASA has $e^x$ out to 2M digits:

$$e^x \approx 2.718281828459045235\cdots$$

$$\epsilon_a = |\epsilon| = 0.00161516179\cdots$$

analytical result $\longrightarrow$ $\epsilon \leq 0.00378\cdots$

# Understanding Error

- An error is a deviation of a result from reality
  - a **systematic error** is introduced by **bias** - this can often be correlated
  - a **random error** is a product of **uncertainty** and impossible to correct
  - random errors can often be **controlled** or **bounded**
- Absolute vs. relative error:

$$\epsilon_a = |\epsilon| = |t(x) - c(x)| \qquad \epsilon_r = \left|\frac{\epsilon_a}{t(x)}\right| = \left|\frac{t(x) - c(x)}{t(x)}\right|$$

- Consider the following true and calculated values:

$$t(x) = 1.5 \quad c(x) = 1.0 \quad \epsilon_a = 0.5 \quad \epsilon_r \approx 0.3\ (30\%)$$

$$t(x) = 100.5 \quad c(x) = 100.0 \quad \epsilon_a = 0.5 \quad \epsilon_r \approx 0.005\ (0.5\%)$$

$$t(x) = 0.03 \quad c(x) = 0.53 \quad \epsilon_a = 0.5 \quad \epsilon_r \approx 16.7\ (1670\%)$$

# Relative Error

- Scientific and engineering applications are less sensitive to small errors in large values
  - $\pm 1$ ton is a lot for a truck but not for a cargo ship
- Relative error can compare values at widely varying scales
- Relative error is often expressed as a **percentage error**: $\epsilon_r \times 100$
- Most computing systems are designed to minimize relative error

- Undefined when the $t(x) = 0$, which can be resolved in a few ways:

**regularization**

$$\epsilon_r = \frac{\epsilon_a}{|t(x)| + \sigma}$$

**specify a value**

$$\epsilon_r = \begin{cases} \frac{\epsilon_a}{|t(x)|} & \text{if } \ t(x) \neq 0 \\ 1 & \text{otherwise} \end{cases}$$

# Relative Error and Interval Measurements

| Fahrenheit | Celsius | Kelvin |
|---|---|---|
| $t(x) = 50.0°\mathrm{F}$ | $t(x) = 10.0°\mathrm{C}$ | $t(x) = 283.0\mathrm{K}$ |
| $c(x) = 51.8°\mathrm{F}$ | $c(x) = 11.0°\mathrm{C}$ | $c(x) = 284.0\mathrm{K}$ |
| $\epsilon_a = 1.8°\mathrm{F}$ | $\epsilon_a = 1°\mathrm{F}$ | $\epsilon_a = 1\mathrm{K}$ |
| $\epsilon_r = 0.036\ (3.6\%)$ | $\epsilon_r = 0.1\ (10\%)$ | $\epsilon_r = 0.0035\ (0.35\%)$ |

- Relative error requires a **ratio scale** with a **natural zero**
- Ratios have to make sense
  - $2x$: there is twice some quantity
  - $\frac{1}{3}x$: there is one third of something
- The zero indicates that there is zero of something